Yating Wu *Ph.D.*

Language, Reasoning, and Reliable AI

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Summary

Research in natural language processing and AI, using discourse structure to improve evaluation and learning. Contributions include identifying interpretation errors missed by standard metrics, predicting question importance, and improving explanation generation.

Selected recognitions: MIT EECS Rising Stars (2025); EMNLP 2024 Outstanding Paper Award (Top 0.4%).

Education

2022 - 2026 **Ph.D. in Electrical and Computer Engineering**, The University of Texas at Austin
Advisors: Junyi Jessy Li, Alexandros G. Dimakis

2020 - 2022 **M.S. in Electrical and Computer Engineering**, The University of Texas at Austin

2014 - 2019 **B.Eng. in Computer Science & B.A. in Japanese**, Dalian University of Technology

2017 - 2018 Exchange student in Computer Science, The University of Tokyo
Advisor: Toshihiko Yamasaki

Work in Progress

Efficient learning from experience
Developing adaptive, dependency-based memory and methods to reduce redundant training data and repeated exploration across agents.

Human feedback and AI safety
Investigating sycophancy and the transfer of safety guardrails across agent environments, focusing on how systems respond to human feedback and follow safety constraints.

AI for scientific discovery
Developing scientific agents that select observations and revise research plans as results become available, with an initial focus on astronomy.

Papers Under Submission

[1] **Yating Wu**, Yuhao Zhang, Sayan Ghosh, Sourya Basu, Anoop Deoras, Jun Huan, Gaurav Gupta "ContextWeaver: Selective and Dependency-Structured Memory Construction for LLM Agents." *Under Submission*, 2026. [Paper]

[2] **Yating Wu**, William Rudman, Venkata S Govindarajan, Alexandros G. Dimakis, Junyi Jessy Li "Multimodal QUD: Inquisitive Questions from Scientific Figures." *Under Submission*, 2026. [Paper]

Publications

** Equal contribution.*

[3] Ramya Namuduri, **Yating Wu**, Anshun Asher Zheng, Manya Wadhwa, Greg Durrett and Junyi Jessy Li "QUDsim: Quantifying Discourse Similarities in LLM-Generated Text." *Conference on Language Modeling* (**COLM**), 2025 [Paper]

[4] **Yating Wu***, Ritika Rajesh Mangla*, Alexandros G. Dimakis, Greg Durrett, Junyi Jessy Li. "Which questions should I answer? Salience Prediction of Inquisitive Questions." *Conference on Empirical Methods in Natural Language Processing* (**EMNLP Oral**), 2024. **Outstanding Paper Award** (Top 0.4%). [Paper]

[5] Negin Raoof*, **Yating Wu***, Carlos Bonilla*, Junyi Jessy Li, Stephanie M Grasso, Alex Dimakis, Zoi Gkalitsiou. "Modeling Bilingual Disfluencies with Large Language Models." *Workshop on LLMs and Cognition in International Conference on Machine Learning* (**ICML Workshop**), 2024. [Paper]

[6] **Yating Wu**, Ritika Rajesh Mangla, Greg Durrett, Junyi Jessy Li. "QUDeval: The Evaluation of Questions Under Discussion Discourse Parsing." *Conference on Empirical Methods in Natural Language Processing* (**EMNLP Oral**), 2023. [Paper]

[7] **Yating Wu***, William Sheffield*, Kyle Mahowald, and Junyi Jessy Li. "Elaborative Simplification as Implicit Questions Under Discussion." *Conference on Empirical Methods in Natural Language Processing* (**EMNLP**), 2023. [Paper]

[8] Wei-Jen Ko, **Yating Wu**, Cutter Dalton, Dananjay Srinivas, Greg Durrett and Junyi Jessy Li. "Discourse Analysis via Questions and Answers: Parsing Dependency Structures of Questions Under Discussion." *Findings of the Association for Computational Linguistics* (**ACL**), 2023. [Paper]

[9] Venelin Kovatchev, Trina Chatterjee, Venkata S Govindarajan, Jifan Chen, Eunsol Choi, Gabriella Chronis, Anubrata Das, Katrin Erk, Matthew Lease, Junyi Jessy Li, **Yating Wu**, Kyle Mahowald. "longhorns at DADC 2022: How many linguists does it take to fool a Question Answering model? A systematic approach to adversarial attacks." *DADC Workshop in The Nations of the Americas Chapter of the Association for Computational Linguistics* (**NAACL Workshop**), 2022 [Paper]

Work Experience

Sept. 2026 - Now **Applied Scientist**, *AWS AI Labs, Santa Clara, CA*
- Training large language models and developing memory and reasoning methods to improve the reliability and efficiency of coding agents

Jun - Oct. 2025 **Applied Scientist Intern**, *AWS AI Labs, Santa Clara, CA*
- Built long-context memory support for LLM coding agents, including dependency-structured context construction for multi-step tool use

May - Aug. 2024 **Applied Scientist Intern**, *Amazon Alexa AGI, Sunnyvale, CA*
- Enhanced planning and tool-use capabilities in LLM agents through question-guided reasoning and agent evaluation

Honors and Fellowships

Honors

2025 **EECS Rising Stars, MIT**

2024 **Outstanding Paper Award, EMNLP 2024**

2021 **1st Place in VMware Codehouse Palo Alto**

2019 **Outstanding Graduates of Dalian University of Technology**

Fellowships

2025 **Outstanding Student Lecture Series Fellowship, UT Austin**

2023-2024 **Professional Development Awards, UT Austin**

2022-2025 **Head TA fellowship, UT Austin**

Teaching Experience

Head Teaching Assistant

CS391L **Machine Learning** (Graduate), Summer 2022 - Fall 2025
Instructors: Adam Klivans, Qiang Liu, UT Austin

Teaching Assistant

CS391L **Machine Learning** (Graduate), Fall 2021 - Spring 2022
Instructors: Adam Klivans, Qiang Liu, UT Austin

Recitation Instructor

EE422C **Software Design and Implementation II** (Undergraduate), Spring 2021
Independently designed and taught weekly recitation sections

EE422C **Software Design and Implementation II** (Undergraduate), Summer 2020
Independently designed and taught weekly recitation sections

Mentoring Experience

Master's student **Ritika Mangla**, UT Austin, 2022-2024. Co-authored two EMNLP papers [4,6], including an Outstanding Paper Award [4].

Master's student **Ashwin Ram**, 2024, UT Austin

Ph.D. student **Tianyang Xu**, TTIC, speech prosody research (work in progress).

Service

Conference Organization

Workflow Chair, EMNLP 2026

Conference Reviewer

ACL Rolling Review (ARR) 2024-Now
ACL 2024, 2025, 2026
EMNLP 2024, 2025, 2026
COLM 2024, 2025, 2026
Workshops: ICML 2025 XAI4Science, AAAI 2025 MARW, NeurIPS 2024 Behavioral ML, ICML 2024 DMLR, ICML 2024 LLMs and Cognition

Department Service

Graduate student representative on faculty hiring panel
TA representative on online course infrastructure building

Presentations

Talks

2025 Which Questions Should I Answer? Salience Prediction of Inquisitive Questions, Outstanding Student Lecture Series, UT Austin

2024 Which Questions Should I Answer? Salience Prediction of Inquisitive Questions, EMNLP 2024

2023 QUDeval: The Evaluation of Questions Under Discussion Discourse Parsing, EMNLP 2023

Poster Presentations

2024 Elaborative Simplification as Implicit Questions Under Discussion, MLL Research Symposium, UT Austin

2023 Elaborative Simplification as Implicit Questions Under Discussion, EMNLP 2023

2023 Discourse Analysis via Questions and Answers: Parsing Dependency Structures of Questions Under Discussion, ACL 2023

Industrial Presentation

2025 ContextWeaver: Selective and Interpretable Long Context Construction for LLM Agents, AWS AI Labs

2024 Self Adaptive Question Generation for Planning Agent, Amazon AGI